SE444: Introduction to Artificial Intelligence

Quiz 3: Bayesian Network Inference

Male Section – Fall 2025 – With Solutions

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Instructions

- This quiz contains 15 multiple choice questions.
- Some questions may have multiple correct answers.
- Topics: Bayesian Inference, Variable Elimination, Factors, d-Separation

Questions and Solutions

Question 1: Inference Problem

Category: Bayesian Inference

When performing probabilistic inference in a Bayesian network to compute $P(X \mid Y = y)$, what role do hidden variables play?

A) They are ignored during computation

B) They must be summed out to marginalize them from the joint distribution

C) They become part of the evidence

D) They replace the query variable

E) They are multiplied to compute the prior

Answer: B

Rationale: Hidden variables appear in the joint but not in the final query/evidence. We marginalize (sum) them out to compute $P(X \mid Y)$.

Question 2: Hidden Variables

Category: Bayesian Inference

In a Bayesian Network with variables {Weather, Traffic, Accident, Delay, Mood} and query $P(\text{Mood} \mid \text{Weather} = \text{rainy}, \text{Accident} = \text{true})$, which are hidden variables?

A) Weather, Accident

B) Traffic, Delay

C) Mood

D) Weather, Traffic, Accident

E) All five variables

Answer: B
Rationale: Query is Mood, evidence is {Weather, Accident}. Hidden variables are those not query or evidence: Traffic and Delay.

Question 3: Enumeration Complexity

Category: Computational Complexity

If a Bayesian network has 6 binary hidden variables, how many total entries need to be summed for inference by enumeration?

A) 6

B) 12

C) 36

D) 64

E) 128

Answer: D
Rationale: For n binary hidden variables, we sum over 2^n combinations. With 6 variables: $2^6 = 64$.

Question 4: Enumeration Formula

Category: Bayesian Inference

In the enumeration formula $P(Q \mid e) = \alpha \sum_h P(Q, e, h)$, what is α?

A) The prior probability of Q

B) A normalization constant that makes probabilities sum to 1

C) The evidence probability

D) The joint probability

E) The number of hidden variables

Answer: B
Rationale: α is the normalization constant: $\alpha = 1/P(e)$, ensuring the resulting probabilities sum to 1.

Question 5: VE vs Enumeration

Category: Variable Elimination

Variable Elimination is more efficient than enumeration because:

A) It uses approximation techniques

B) It computes intermediate results (factors) once and reuses them

C) It ignores less important variables

D) It converts the network to a tree

E) It only works with binary variables

Answer: B
Rationale: VE avoids redundant computation by computing factors once and reusing them, like factoring in algebra before computing.

Question 6: Factor Definition

Category: Variable Elimination

A CPT $P(C \mid A, B)$ in Variable Elimination is represented as:

A) A factor $f(A, B, C)$ containing probability values for each combination

B) A single probability value

C) An edge from A to C and B to C

D) The product $P(A) \times P(B) \times P(C)$

E) A normalization constant

Answer: A
Rationale: CPTs are factors—multi-dimensional tables mapping each assignment of (A, B, C) to probability values.

Question 7: Factor Operations

Category: Variable Elimination

When eliminating variable X from VE, which operation is performed first?

A) Marginalization over X

B) Factor product of all factors containing X

C) Normalization

D) Factor division

E) Setting X to evidence

Answer: B
Rationale: To eliminate X: (1) First multiply all factors mentioning X, then (2) sum out X from the product.

Question 8: Marginalization Result

Category: Variable Elimination

If you have factors $f(A, B)$ and $g(B, C)$ and you want to eliminate B, what is the scope of the resulting factor?

A) Only A

B) Only B

C) Only C

D) A and C

E) A, B, and C

Answer: D
Rationale: Multiply: $f(A, B) \times g(B, C) = h(A, B, C)$. Then marginalize B: $\sum_B h(A, B, C) = h'(A, C)$.

Question 9: VE Computation

Category: Variable Elimination

In a network $D \to E \to F$, to compute $P(F \mid D = \text{true})$, which variable is the hidden variable that must be eliminated?

A) D

B) E

C) F

D) Both D and F

E) None, direct computation possible

Answer: B
Rationale: Query is F, evidence is D. E is the hidden variable that must be eliminated by summing over its values.

Question 10: Numerical VE

Category: Computation

Given $P(X)$ with $P(X = T) = 0.4$, and $P(Y \mid X)$ with $P(Y = T \mid X = T) = 0.9$ and $P(Y = T \mid X = F) = 0.1$. What is $P(X = F, Y = T)$?

A) 0.04

B) 0.06

C) 0.36

D) 0.40

E) 0.54

Answer: B
Rationale: $P(X = F, Y = T) = P(X = F) \times P(Y = T \mid X = F) = 0.6 \times 0.1 = \boxed{0.06}$. This is factor product.

Question 11: Normalization

Category: Computation

VE produces unnormalized results: $P(X = T, e) = 0.12$ and $P(X = F, e) = 0.28$. What is $P(X = F \mid e)$?

A) 0.12

B) 0.28

C) 0.30

D) 0.40

E) 0.70

Answer: E
Rationale: Normalize: $P(X = F \mid e) = \frac{0.28}{0.12+0.28} = \frac{0.28}{0.40} = \boxed{0.70}$.

Question 12: d-Separation Inference

Category: Conditional Independence

In a chain $X \to Y \to Z$, what does d-separation tell us about inference with evidence $Y = y$?

A) X and Z become dependent

B) X and Z become conditionally independent given Y

C) Y can be removed from the network

D) X directly causes Z

E) Inference requires all three variables

Answer: B
Rationale: In a chain, the middle node blocks information flow. Given Y, $X \perp Z \mid Y$ so $P(X \mid Z, Y) = P(X \mid Y)$.

Question 13: Elimination Order

Category: Variable Elimination

What happens if you choose a poor elimination order in Variable Elimination?

A) The result becomes incorrect

B) Intermediate factors may become exponentially large, slowing computation

C) The network structure changes

D) Evidence values are lost

E) Hidden variables cannot be identified

Answer: B
Rationale: Poor ordering creates large intermediate factors. The result is always correct, but efficiency suffers dramatically.

Question 14: Critical Thinking

Category: Critical Thinking

A screening test has: $P(\text{Disease}) = 0.02$, $P(\text{Positive} \mid \text{Disease}) = 0.95$, $P(\text{Positive} \mid \neg\text{Disease}) = 0.10$. A patient tests positive. What is $P(\text{Disease} \mid \text{Positive})$?

A) About 16%

B) About 35%

C) About 50%

D) About 95%

E) Cannot be determined

Answer: A
Rationale: Using Bayes' theorem:

$$P(D \mid +) = \frac{P(+ \mid D) \cdot P(D)}{P(+ \mid D) \cdot P(D) + P(+ \mid \neg D) \cdot P(\neg D)} = \frac{0.95 \times 0.02}{0.95 \times 0.02 + 0.10 \times 0.98} = \frac{0.019}{0.117} \approx \boxed{16.2\%}$$

Key Insight: Low prevalence dominates—most positives are false positives!

Question 15: VE Properties

Category: Variable Elimination

Which of the following is **FALSE** about Variable Elimination?

A) VE always produces exact inference results

B) VE works on any directed acyclic graph

C) VE complexity depends on elimination order

D) VE is guaranteed polynomial time for all networks

E) VE uses factor product and marginalization operations

Answer: D

Rationale: D is FALSE. VE can be exponential for networks with high treewidth. All other statements are true.

Answer Key Summary

Q#	Topic	Answer
1	Inference Problem	B
2	Hidden Variables	B
3	Enumeration Complexity	D
4	Enumeration Formula	B
5	VE vs Enumeration	B
6	Factor Definition	A
7	Factor Operations	B
8	Marginalization Result	D
9	VE Computation	B
10	Numerical VE	B
11	Normalization	E
12	d-Separation Inference	B
13	Elimination Order	B
14	Critical Thinking	A
15	VE Properties	D